

DualVoice: A Speech Interaction Method Using Whisper-Voice as Commands

Jun Rekimoto
The University of Tokyo
Bunkyo-ku, Tokyo, Japan
Sony CSL Kyoto
Kyoto-shi, Kyoto, Japan
rekimoto@acm.org

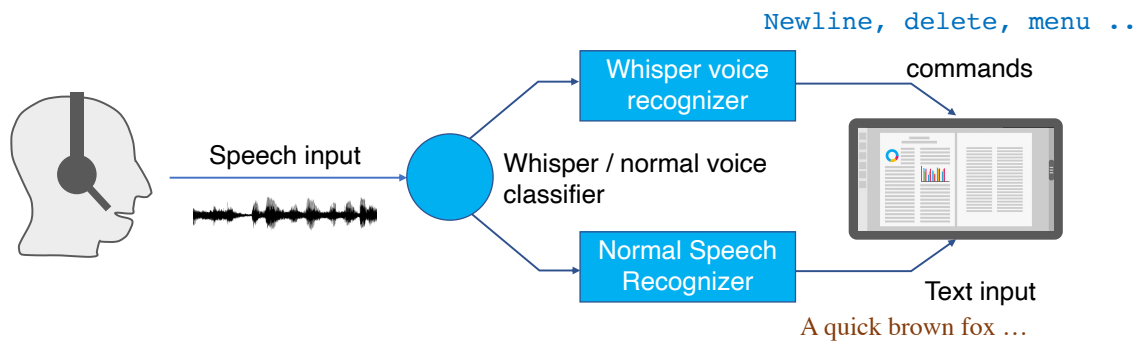


Figure 1: DualVoice speech interaction: DualVoice uses “whispering” for commands while a normal voice is used for text inputs.

ABSTRACT

Applications based on speech recognition have become widely used, and speech input is increasingly being utilized to create documents. However, since there is no easy way to distinguish commands and text input in speech, it is still difficult to correct misrecognition by speech, which makes it necessary to re-edit documents by manual input. It is also difficult to input symbols and commands because these may be misrecognized as text letters. We propose a speech interaction method called DualVoice, in which commands are input in a whispered voice and letters in a normal voice. The proposed method does not require any special hardware other than a regular microphone, thus enabling a complete hands-free interaction. It can be used in a wide range of situations where speech recognition is already available. We designed two neural networks, one for discriminating normal speech from whispered speech, and the second for recognizing whisper speech.

CCS CONCEPTS

• **Human-centered computing** → Sound-based input / output; Interface design prototyping; Mobile devices; • **Computing methodologies** → Neural networks.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519700>

KEYWORDS

speech interaction, whisper voice recognition, whisper voice classification, neural networks

ACM Reference Format:

Jun Rekimoto. 2022. DualVoice: A Speech Interaction Method Using Whisper-Voice as Commands. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3519700>

1 INTRODUCTION

Automatic speech recognition (ASR) is being used in a variety of applications ranging from the operation of devices such as smart speakers or car navigation systems to techniques for text input. Voice input does not require any special input device other than a microphone and can be operated hands-free. Furthermore, as text input based on speech can be much faster than typing, voice input is also used to jot down ideas and quickly enter manuscript drafts.

However, with speech recognition there are problems with recognition errors. When speech recognition is used for text input, recognition errors must be corrected using the keyboard or mouse, which cancels out the advantage of hands-free operation. In addition, when using speech to input text, problems can arise when inputting commands or special characters other than regular text. For example, if you want to enter ‘?’, you may say ‘question mark’, but it might be entered as the text itself, not the symbol. If you want to start a new paragraph, and you say ‘new paragraph’, it may be recognized literally and input as the text ‘new paragraph’. Similar problems occur when inputting commands such as ‘delete word’ and ‘new paragraph’, which are frequently used in text editing.

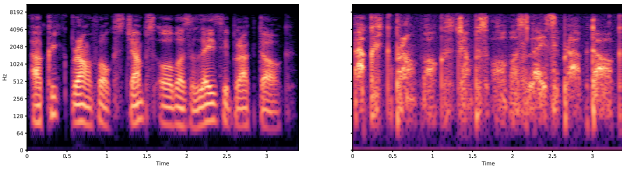


Figure 2: Spectrograms of normal and whispered voices saying “A quick brown fox jumps over the lazy black dog.”

It might be possible to decide in advance that saying ‘new line’ always means the new-line command and not text input, but in that case, when you want to enter the string ‘new line’, you will be in trouble. Moreover, it would be a burden for users to remember all the commands and text input phrases. The shift button could be used to switch between modalities, but the hands-free advantage of voice input would be lost.

To address these issues, we propose a voice input method called DualVoice that uses “whispering” for commands and uses a normal voice for text inputs (Fig. 1). Whispering is a mode of speech that many users can utter without any special training. With whisper and normal voice, we can give multiple meanings to speech. For example, saying ‘new line’ in a normal voice means to input that string, but whispering ‘new line’ means to invoke a new-line command. To correct a recognition error, users can whisper ‘candidates’ to show alternative recognition candidates and select one of them by whispering the number (‘one’, ‘two’, etc.) attached to that candidate.

The relationship between normal voice and whispering voice can be compared to the relationship between text input and command input. The conventional voice recognition methods can be regarded as keyboard that did not include command or symbol keys. However, a keyboard used for computer interactions provides function keys, command keys, keys for symbols, and other means of entering commands, as well as letter input. Thus, we can invoke those functions without explicitly changing the interaction mode or having to take our hands off the keyboard. Similarly, with the proposed DualVoice method, text and command input can co-exist without any special mode conversion operations.

Figure 2 shows spectrograms of utterances with normal and whispered voices. To realize DualVoice interaction, we have developed two types of neural networks: one to distinguish between whispered and normal voices, and the other to recognize whispered utterances.

2 RELATED WORK

Goto et al. have proposed ‘speech-shift’, which specifies the mode of speech input by intentionally controlling the pitch of speech [8]; if the F0 of an utterance exceeds a specified threshold, it is judged as a different mode. This method is close to the purpose of this paper, but it forces the user to speak at an unnaturally high pitch for stable recognition. Switching between whisper and normal speech can be done more clearly without setting the (user-unknown) threshold.

Goto et al. also developed a method that automatically detects filled (vocalized) pauses, which are one of the hesitation phenomena

of utterances in speech commands, and then proposes possible candidates that can fill the command [7]. For instance, if the user says “play, Beeee...” and stops, the system detects “eeee” as the filled vocalized pauses of hesitation and proposes filling candidates such as “Beatles” or “Beach Boys”. This method demonstrates the possibility of indicating non-verbal intentions in speech, but it can only utilize hesitation, not arbitrary commands as our method does. Moreover, making vocalized pauses is only possible after vowels in speech, not after consonants.

DualBreath [15] is a system that uses breathing as a command, distinguishing it from normal exhalation by discriminating between inhaling and exhaling air through the nose and mouth simultaneously. This system can express triggers equivalent to a simple button press, but it cannot express commands as rich as whispers. ProxiMic is a sensing technology that detects a user’s utterance by a microphone device located close to the mouth [19]. It is intended for use as a wake-up-free speech for smart speakers and similar voice-controlled devices. However, it requires physical motion (moving a microphone close to the mouth), and thus mixing normal and close-to-the-mouth utterances is not easy.

Fukumoto proposed SilentVoice, where a user can input speech by using “ingressive speech”, an utterance made while inhaling (breathing-in) [5]. Although the method is designed mainly for silent speech, it can also distinguish ingressive speech from normal speech. However, it requires a special microphone placed very close to the mouth, and training is required for users to correctly speak in the ingressive mode. Moreover, frequently changing between normal and ingressive utterances is difficult.

With some Speech Input Systems, such as Google Cloud Speech-to-Text [11], you will get ‘.’ input when you say “period”. However, in order to distinguish this utterance from the ‘period’ that appears in the sentence, it is necessary to leave a certain amount of pause before and after it, which reduces the overall input speed.

There also have been researches on whisper voice recognition [3, 6, 9]. However, use of mixing normal voices and whispering was not considered.

The Alexa smart speaker supports *whisper mode* [10]. When this mode is set, if you talk to Alexa in a whisper, it will respond in a whisper [4].

3 INTERACTING WITH DUALVOICE

Figure 3 shows the examples of voice input process using DualVoice. In DualVoice, normal voice input is converted into text, the same as in conventional voice input text creation, while in contrast, symbols such as ‘COMMA’, ‘PERIOD’, and ‘QUOTE’ are input as symbols when spoken in a whisper. Whispering ‘NEW LINES’ is also treated as a command, and means making a new line. If there is a misrecognition, it is possible to delete the last word or change the word of interest by whispering ‘BACK’ or ‘DELETE SENTENCE’. It is also possible to present an alternative recognition candidate by whispering ‘MENU’. Candidates on a menu are labeled as 1, 2, 3 and can be selected by whispering ‘ONE’, ‘TWO’, ‘THREE’, etc. (Fig. 3)

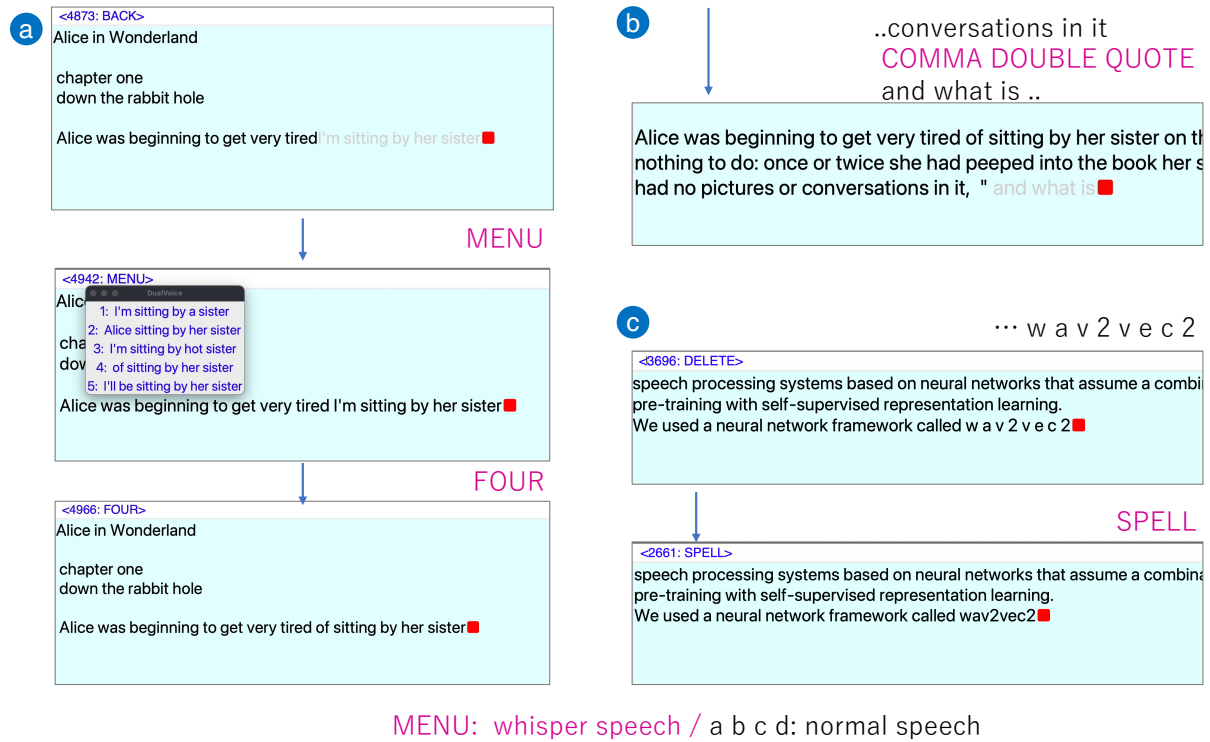


Figure 3: Examples of DualVoice Interactions: (a) Correction of recognition errors: whispering 'MENU' invokes a menu for possible recognition candidates. A user can select it by whispering numbers. (b) Entering symbols by whispering 'COMMA' and 'DOUBLE QUOTE'. and (c) combined the spelled input by whispering 'SPELL'.

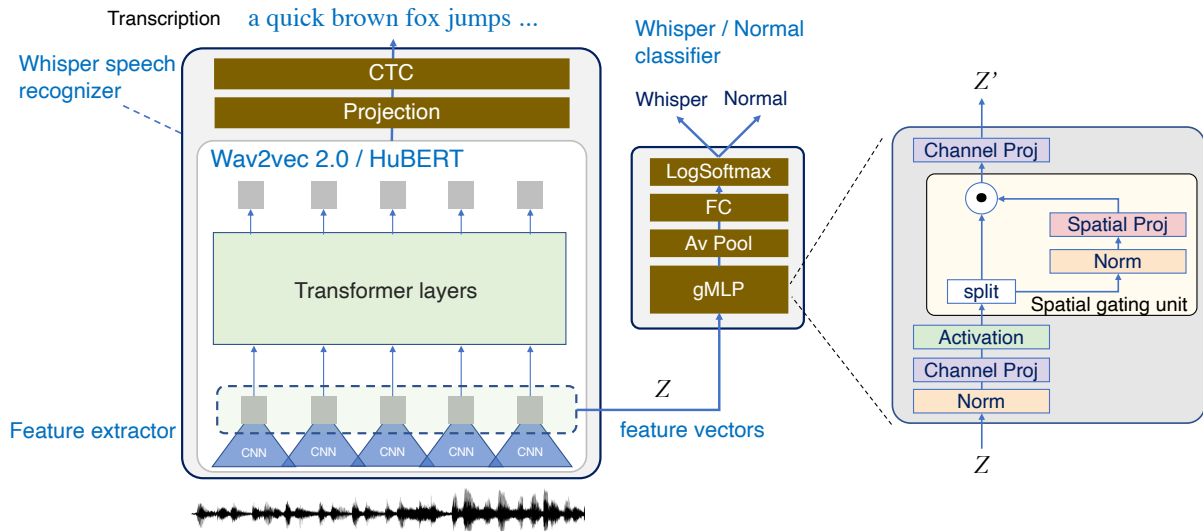


Figure 4: Whisper voice recognition and whisper / normal classifier.

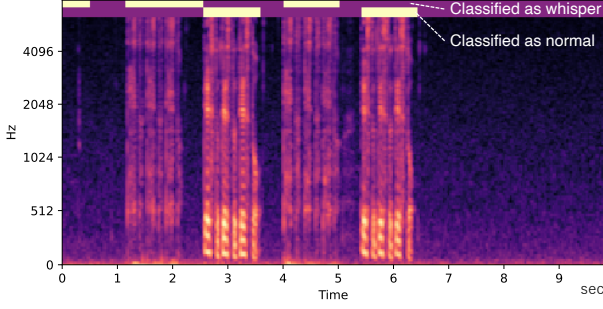


Figure 5: Whisper classification results.

4 RECOGNITION ARCHITECTURE

4.1 Whisper Voice Recognition

For whispered voice recognition, we use wav2vec 2.0 [1, 22], which is a self-supervised neural networks designed for speech processing system. It is a speech processing systems based on neural networks that assume a combination of pre-training with self-supervised representation learning on unlabeled speech data and fine-tuning on labeled speech data. These systems are mainly intended for speech recognition applications but have also been applied to speaker recognition, language recognition, and emotion recognition [18, 22].

4.2 Whisper Voice Classification

The whisper voice classification part distinguishes whispers from normal voice input by a fixed-length (e.g., 100 ms) audio signal (Fig. 5). The feature extractor obtained from wav2vec 2.0 converts the acoustic signal into 512-dimensional features every 20ms (Fig. 4). Using these features, the Gated Multi-Layer Perception (gMLP) layer [13] is applied, followed by the average pooling layer, and the Fully-Connected (FC) layer (Fig. 4). It is trained as a binary classifier of normal and whisper voices.

The gMLP proposed by [13] is one of the simpler (and potentially better) alternatives to Transformer-type neural networks. It receives $Z \in \mathbb{R}^{n \times \text{dim}}$ feature vectors (where dim denotes the dimension of the feature vector and n is the number of vectors along the time dimension) and generates the same size vectors Z' . In our case, n is fixed to 4 because the input audio length is fixed to 100ms, and dim is 512, the same as the feature extractor's output dimension.

The details of gMLP are also shown in Fig. 4. The spatial gating unit (SGU) block calculates

$$\begin{aligned}\tilde{Z} &= \delta(ZU) \\ s(\tilde{Z}) &= \tilde{Z} \odot (W\tilde{Z} + b) \\ Z' &= s(\tilde{Z})V\end{aligned}$$

where $U \in \mathbb{R}^{\text{dim} \times \text{internaldim}}$ and $V \in \mathbb{R}^{\text{internaldim} \times \text{dim}}$ are linear projections along the channel dimension, δ is an activation function, and $\tilde{Z} \in \mathbb{R}^{n \times \text{internaldim}}$ denotes projected feature vectors. In our case, internaldim is fixed to 3072. $\odot$ refers to element-wise multiplication, $W \in \mathbb{R}^{n \times n}$ is a matrix for which the size is the same as the sequence length, and b refers to a bias term. This equation

one, two, three, four, five	six, seven, eight, nine, zero
A, B, C, D, E, F, G	H, I, J, K, L, M, N
O, P, Q, R, S, T, U	V, W, X, Y, Z
newline, back, delete	delete sentence
space, equal, at, number	dollar, ampersand, asterisk
left parenthesis, right parenthesis	left bracket, right bracket
underline, hyphen, plus, minus	percent, atmark, sharp
spell, paragraph	period, comma, dot
menu, open, close, yes, no	line, new, repeat, candidates
next, page, word	delete line, delete word
question mark, exclamation mark	quote, double quote

Table 1: Whisper phrases used for per-user training.

is intended to modulate individual feature vector representations over time series.

4.3 Training

For training, we start from the neural networks pre-trained and fine-tuned with (normal) voice data (960 hours from the Librispeech dataset [16]). Note that the pre-training does not require the corresponding text, only the voice data.

Then, we further fine-tune with the following two whispered voice datasets.

wTIMIT wTIMIT A whisper voice corpus (*whisper TIMIT*) [12].

Each speaker both says and whispers 450 phonetically balanced sentences according to the prompts in TIMIT. There are 29 speakers and they are somewhat gender-balanced. The total number of utterances we used is 11,324 (1,011 minutes). The data is provided in two parts, train and test. We will use this division as is.

Per-user dataset Voice data dubbed with whisper voice by each user; sequences of selected voice commands are used as scripts (Table 1). These phrases are mainly assumed to be the commands used during text input. Each user repeats each phrase five times, and a total of 110 phrases are recorded. These phrases are further randomly concatenated to be used as a dataset. After this concatenation, the total number of utterances is 936 (about 82 minutes).

We apply fine-tuning in two stages. In the first stage, we train on the wTIMIT (whisper voice), and in the second stage, we use the whispering voice command set blown by the users (per-user dataset).

We also use wTIMIT normal and whisper voice data as the training dataset for the training of whisper classification. The length of the voice supplied to the classifier is set to 1600 samples (100ms in length with 16K/sec audio sampling). This is consistent with the length of the speech chunks used in the speech recognition cloud service in the later stage.

To identify the whispering voice, we used the fixed feature extractor part of wav2vec 2.0 after pre-training (Fig. 4). This was so that the whisper classification could partially share the neural network with the whisper recognition, thus reducing the overall network size. The output of the feature extractor was passed

model	train	test	WER (%)	CER (%)
wav2vec 2.0 large [2]	LS-960	wTIMIT(N)	15.85	4.42
	LS-960	wTIMIT(W)	39.00	16.46
	LS-960 + wTIMIT(W)	wTIMIT(W)	0.84	0.45
	LS-960	per-user	83.95	38.42
	LS-960 + wTIMIT(W)	per-user	53.33	34.64
	LS-960 + per-user	per-user	2.45	0.56
	LS-960 + wTIMIT(W) + per-user	per-user	0.38	0.18

Table 2: Whisper voice recognition accuracy for wav2vec2 large. The model is pre-trained on unlabeled Librispeech 960 hours (LS-960) normal voice data and fine-tuned on the same LS-960 with text labels. The ‘train’ column indicates dataset used for fine-tuning (LS-960: Librispeech 960 hours, wTIMIT(N): wTIMIT normal voice, wTIMIT(W): wTIMIT whisper voice, and per-user: per-user whisper voice).

model	accuracy
pooling + FC + FC	0.87
gMLP (6 layers) + pooling + FC	0.96
gMLP (1 layer) + pooling + FC	0.96

Table 3: Whisper-normal classification accuracy of a simple pooling + FC (fully connected) model and a model with gMLP. The latter achieved better recognition accuracy. Trained and tested with the wTIMIT normal and whisper dataset.

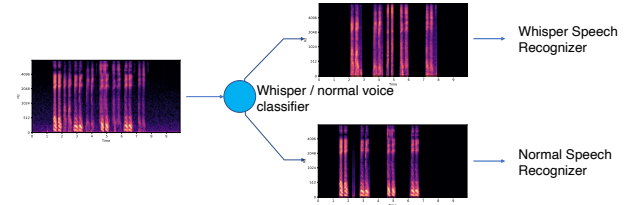


Figure 6: Whisper classification and filtering

through gMLP [13], the average pooling layer, the fully connected layer, and the LogSoftmax layer to obtain binary logits.

5 RESULTS

5.1 Whisper voice recognition:

The results of the training are shown in Table 2. The recognition scores are shown as word error rate (WER) and character error rate (CER). The training times (using nVIDIA A6000) for fine-tuning were 4.5 hours for wTIMIT(W) and 20 minutes for the per-user dataset.

We found that the recognition accuracy was poor when trained (fine-tuned) with LS-960 and wTIMIT(W) and tested with the per-user dataset. This result means that this neural network is not sufficient for an unspecified speaker whisper recognizer. In contrast, the system pre-trained on LS-960, and fine-tuned with LS-960, wTIMIT(W), and per-user, showed good recognition accuracy (e.g., WER 0.38%).

These results lead us to conclude that (1) the model pre-trained on normal speech can deliver a good whisper recognition accuracy when it is fine-tuned with whisper voice, and (2) at this stage, only the user-dependent whisper recognition is achievable.

5.2 Whisper-Normal Classification:

For the whisper-normal classifier training, we used the normal and whisper voices of wTIMIT. The 16-kHz sampled voice data were divided into an audio segment containing 16,000 samples (100 msec). Each segment was labeled with either normal or whisper. The system was then trained to discriminate between these two.

Of the 100-msec speech segments, those with an average power of less than $-20db$ were excluded, and training was performed on the rest. The results are shown in Table 3). As shown in the table, we were able to classify normal and whispered speech with a 96% accuracy. We also compared the accuracy of gMLP with 6-layers and gMLP with 1-layer, and could not see no significant difference. We thus use gMLP 1-layer as our whisper classifier.

5.3 Normal Voice Recognition

For normal voice recognition, although we could train the same neural network described above, we opted instead to select an existing cloud-based speech recognition system (Google Cloud Speech-to-Text [11]) because of its better recognition accuracy and compatibility with existing voice-based text input applications.

6 IMPLEMENTATION DETAILS

The neural networks for whisper recognition and whisper discrimination are based on huggingface [21] wav2vec2 classe, and other networks developed with PyTorch [17]. We use the PyTorch gMLP implementation of [20]. The GUI system for the DualVoice text input example is built using the Tkinter Python GUI platform [14].

The GUI system manages the thread that receives the microphone input and splits it into packets containing 1,600 samples (100ms) of sound. These packets are sent via TCP/IP to the whisper discriminator, which discriminates between whisper, normal, and silence for each packet. In the case of speech sent to Whisper speech recognition, packets that are judged not to be Whisper speech will be replaced with no-speech packets; in the case of Normal speech recognition, packets that are judged to be Whisper speech will be replaced with no-speech packets (Fig. 6).

7 DISCUSSIONS

7.1 Other Potential Applications

There are more applications of whisper voice than just text input. For example, if you want to have multiple virtual avatars speak, you can switch between having avatar 1 speak with a normal voice and avatar 2 speak with a whisper voice.

When this technology is incorporated into wearable computers, the normal voice will be used for normal conversations and the whisper voice will be used for conversations with the wearable computer.

During a remote conference, the whisper voice could be used as a command to the computer, and that part could be muted.

7.2 Combination with Silent Speech

In this paper, we utilized whisper voice and normal voice to represent multiple modalities of interaction. In the same way, it should also be possible to use silent speech and normal speech, although other sensors than a microphone are required for silent speech recognition. Further, there is potential to use three modalities: silent speech, normal speech, and whispered speech.

7.3 User-Independent Whisper Recognition

In the implementation presented in this study, we use an existing speech recognition system (Google Cloud Speech-to-Text [11]) for normal voice recognition and our custom neural network for whisper voice recognition. The former is trained on a large corpus, is speaker-independent, and can be used without special training. In contrast, the whisper speech recognition part is trained on a smaller corpus and requires a training phase in which individual users are asked to read out example sentences in a whisper for speaker adaptation.

As whisper-based interactions become more widespread, we foresee that it will become possible to obtain a whisper-speech dataset from the usage history. As a result, we will be able to use whisper speech recognition in a speaker-independent manner.

8 CONCLUSION

We proposed DualVoice, a speech input method for inputting non-text commands in a whispered voice and inputting text in a normal voice. While normal voice input is used for text input, various commands can be entered by whispering them. Our method does not require any special hardware other than a regular microphone and can be used in a wide range of situations where speech recognition is already available. We designed two neural networks, one for distinguishing whisper speech from normal speech and the other for recognizing whisper speech. We then implemented a prototype speech-based text input system using these neural networks and evaluated its usability.

ACKNOWLEDGMENTS

This work was supported by JST Moonshot R&D Grant Number JPMJMS2012, JST CREST Grant Number JPMJCR17A3, and JSPS KAKENHI Grant Number 19H04148.

REFERENCES

- [1] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. *arXiv [cs.CL]* (June 2020).
- [2] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. *arXiv [cs.CL]* (June 2020).
- [3] Heng-Jui Chang, Alexander H Liu, Hung-Yi Lee, and Lin-Shan Lee. 2020. End-to-end Whispered Speech Recognition with Frequency-weighted Approaches and Pseudo Whisper Pre-training. (May 2020). *arXiv:2005.01972 [cs.CL]*
- [4] Marius Cotesco, Thomas Drugman, Goeric Huybrechts, Jaime Lorenzo-Trueba, and Alexis Moinet. 2019. Voice Conversion for Whispered Speech Synthesis. (Dec. 2019). *arXiv:1912.05289 [cs.SD]*
- [5] Masaaki Fukumoto. 2018. SilentVoice: Unnoticeable Voice Input by Ingressive Speech. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology* (Berlin, Germany) (UIST '18). Association for Computing Machinery, New York, NY, USA, 237–246.
- [6] Shabnam Ghaffarzadegan, Hynek Boril, John H L Hansen, Shabnam Ghaffarzadegan, Hynek Boril, John H L Hansen, Shabnam Ghaffarzadegan, John H L Hansen, and Hynek Boril. 2016. Generative Modeling of Pseudo-Whisper for Robust Whispered Speech Recognition. *IEEE/ACM Trans. Audio, Speech and Lang. Proc.* 24, 10 (Oct. 2016), 1705–1720.
- [7] Masataka Goto. 1999. A Real-time Filled Pause Detection System for Spontaneous Speech Recognition. *Proc. of Eurospeech '99* (1999).
- [8] Masataka Goto, Yukihiro Omoto, Katunobu Itou, and Tetsunori Kobayashi. 2003. Speech shift: direct speech-input-mode switching through intentional control of voice pitch. In *8th European Conference on Speech Communication and Technology, EUROSPEECH 2003 - INTERSPEECH 2003, Geneva, Switzerland, September 1-4, 2003*.
- [9] Dordé T Grozdic and Slobodan T Jovicic. 2017. Whispered Speech Recognition Using Deep Denoising Autoencoder and Inverse Filtering. *IEEE/ACM Trans. Audio, Speech and Lang. Proc.* 25, 12 (Dec. 2017), 2313–2322.
- [10] Amazon.com Inc. 2018. How Alexa keeps getting smarter. <https://www.aboutamazon.com/devices/how-alexa-keeps-getting-smarter>
- [11] Google Inc. 2020. Google Cloud Speech-to-Text. <https://cloud.google.com/speech-to-text>
- [12] Boon Pang Lim. 2010. *Computational differences between whispered and non-whispered speech*. Ph.D. Dissertation. University of Illinois Urbana-Champaign.
- [13] Hanxiao Liu, Zihang Dai, David R So, and Quoc V Le. 2021. Pay Attention to MLPs. (May 2021). *arXiv:2105.08050 [cs.LG]*
- [14] Fredrik Lundh. 1999. An introduction to tkinter. www.pythonware.com/library/tkinter/introduction/index.htm (1999).
- [15] Ryoya Onishi, Tao Morisaki, Shun Suzuki, Saya Mizutani, Takaaki Kamigaki, Masahiro Fujiwara, Yasutoshi Makino, and Hiroyuki Shinoda. 2021. DualBreath: Input Method Using Nasal and Mouth Breathing. In *Augmented Humans Conference 2021* (Rovaniemi, Finland) (AHs'21). Association for Computing Machinery, New York, NY, USA, 283–285.
- [16] Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. 2015. Librispeech: An ASR corpus based on public domain audio books. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 5206–5210. <https://doi.org/10.1109/ICASSP.2015.7178964>
- [17] Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan, Edward Yang, Zachary DeVito, Zeming Lin, Alban Desmaison, Luca Antiga, and Adam Lerer. 2017. Automatic differentiation in PyTorch. (2017).
- [18] Leonardo Pepino, Pablo Riera, and Luciana Ferrer. 2021. Emotion Recognition from Speech Using Wav2vec 2.0 Embeddings. (April 2021). *arXiv:2104.03502 [cs.SD]*
- [19] Yue Qin, Chun Yu, Zhaozheng Li, Mingyuan Zhong, Yukang Yan, and Yuanchun Shi. 2021. ProxiMic: Convenient Voice Activation via Close-to-Mic Speech Detected by a Single Microphone. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–12.
- [20] Phil Wang. 2021. pytorch-gMLP: Implementation of gMLP, an all-MLP replacement for Transformers, in Pytorch. <https://github.com/lucidrains/g-mlp-pytorch>.
- [21] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M Rush. 2019. HuggingFace's Transformers: State-of-the-art Natural Language Processing. (Oct. 2019). *arXiv:1910.03771 [cs.CL]*
- [22] Cheng Yi, Jianzhong Wang, Ning Cheng, Shiyu Zhou, and Bo Xu. 2020. Applying Wav2vec2.0 to Speech Recognition in Various Low-resource Languages. (Dec. 2020). *arXiv:2012.12121 [cs.CL]*